

Module 6 Week B — Stretch: Embedding Space Explorer

Honors Track. This is an Honors Track stretch assignment. Stretch assignments are for learners who have completed all core assignments, are On Track or Advanced, and are attending consistently. If you are behind on core work, focus there first.

Due: Thursday end of day. Resubmissions are accepted through Saturday of the assignment week.

Prerequisite: None — this stretch builds on base lab work from Week B.

The Challenge

Word embeddings and document embeddings live in high-dimensional space that humans cannot directly visualize. But dimensionality reduction techniques (t-SNE, Principal Component Analysis (PCA)) can project these vectors into 2D, revealing the semantic structure that embeddings capture.

Build a visualization tool that makes embedding space visible. Project word-level GloVe vectors and document-level DistilBERT embeddings into 2D, and create annotated scatter plots that reveal how embeddings organize meaning.

This is a standard NLP analysis technique. Embedding visualizations are used in production to debug retrieval systems, validate embedding quality, and communicate model behavior to non-technical stakeholders.

What You'll Learn

- Dimensionality reduction with t-SNE and/or PCA (`sklearn.manifold.TSNE` , `sklearn.decomposition.PCA`)
- Visualizing high-dimensional structure in 2D scatter plots
- Interpreting embedding clusters: what semantic relationships survive dimensionality reduction
- Comparing word-level and document-level embedding spaces
- Creating annotated, publication-quality visualizations with matplotlib

Outcome

Your deliverable extends your Lab 6B repo (`m6-l6b-{your-username}`), the GitHub Classroom repo from the lab — `lab-6b-embeddings` is the lab's working branch, not the repo name) and adds:

Artifact	Description
Word embedding visualization	2D scatter plot of 200 GloVe vectors, color-coded by semantic category
Document embedding visualization	2D scatter plot of 20 BBC News article DistilBERT embeddings, color-coded by article category
1-paragraph analysis	What the embedding space reveals about the relationships between your chosen word categories and the BBC News topics

Constraints:

- Select 200 words from the GloVe vocabulary and organize them into at least 5 semantic categories of your choice (e.g., countries, sports, business/finance terms, technology, emotions — the categories are your design choice; justify them briefly)
- Apply t-SNE or PCA (your choice – document why) to reduce the 200 GloVe vectors (50d) to 2D
- Create a scatter plot with clear color coding by category, a legend, and annotations for at least 10 notable words
- Select 20 articles from `data/bbc_news.csv` with at least 3 per category — pick from across all 5 BBC News categories (business, entertainment, politics, sport, tech). Compute DistilBERT embeddings. Apply the same reduction technique to 2D
- Create a second scatter plot color-coded by BBC News category
- The analysis must reference specific clusters, outliers, or patterns visible in your plots

Tips

- t-SNE is better at preserving local structure (nearby points stay nearby). PCA is better at preserving global structure (distant points stay distant). For word clusters, t-SNE usually produces more visually interpretable results. For document-level comparisons, PCA may be clearer.
- t-SNE has a `perplexity` parameter that controls the balance between local and global structure. Try values between 5 and 50 for 200 points. Low perplexity emphasizes tight clusters; high perplexity shows broader patterns.
- If your word scatter plot is too crowded, annotate only the most interesting words (cluster centers, outliers, words that appear in unexpected clusters).
- For the document plot, use `plt.annotate` with short text labels (first 30 characters of each article, or the article's category) so you can identify which articles cluster together.

Submission

1. In your Lab 6B repo (`m6-l6b-{your-username}`), create a new branch named `stretch-embedding-explorer` off `main`
2. Add your visualization script (e.g., `stretch_embedding_explorer.py`), the saved plots, and your written analysis (e.g., `stretch_analysis.md`)
3. Push the branch and **open a separate pull request from `stretch-embedding-explorer` to `main`** — do not add these commits to your existing Lab 6B PR. The stretch PR can stay open; it will not affect the lab autograder
4. Paste the **stretch PR URL** (not the lab PR URL, not the bare repo URL) into TalentLMS -> Module 6 Week B -> Stretch 6B-S1